Tutorials Exercises W3Schools app iOS & Android NEW Sign In
. CSS JAVASCRIPT SQL PYTHON JAVA PHP C C++ C# AWS

MySQL PRIMARY KEY Constraint

‹ Previous Next ›

SHARE

MySQL PRIMARY KEY Constraint

The `PRIMARY KEY` constraint uniquely identifies each record in a database table.

A `PRIMARY KEY` constraint ensures unique values, and cannot contain NULL values (it is a combination of both a `UNIQUE` constraint and a `NOT NULL` constraint).

A table can have only ONE `PRIMARY KEY` constraint. The primary key can either be a single column, or a combination of columns.

Tip: The primary key is the target for `FOREIGN KEY` constraints in other tables (which enforces referential integrity between data in two tables).

PRIMARY KEY on CREATE TABLE

Tutorials Exercises W3Schools app iOS & Android NEW Sign In
. CSS JAVASCRIPT SQL PYTHON JAVA PHP C C++ C# AWS

```
CREATE TABLE Persons (
    ID int PRIMARY KEY,
    LastName varchar(255) NOT NULL,
    FirstName varchar(255),
    Age int
);
```

PRIMARY KEY on Multiple Columns

To define an **un-named** `PRIMARY KEY` constraint on multiple columns, use the following SQL syntax:

```
CREATE TABLE Persons (
    ID int,
    LastName varchar(255),
    FirstName varchar(255),
    Age int,
    PRIMARY KEY (ID, LastName)
);
```

Note: In the example above, the `PRIMARY KEY` value is made up of two columns (ID + LastName).

To define a **named** `PRIMARY KEY` constraint on multiple columns, use the following SQL syntax:

```
CREATE TABLE Persons (
    ID int,
    LastName varchar(255),
    FirstName varchar(255),
    Age int,
```

Tutorials Exercises W3Schools app iOS & Android NEW Sign In
. CSS JAVASCRIPT SQL PYTHON JAVA PHP C C++ C# AWS

Note: In the example above, the `PRIMARY KEY` is named "PK_Person", and the value is made up of two columns (ID + LastName).

REMOVE ADS

PRIMARY KEY on ALTER TABLE

To create a `PRIMARY KEY` constraint on the "ID" column when the table already has been created, use the following SQL:

```
ALTER TABLE Persons
ADD PRIMARY KEY (ID);
```

PRIMARY KEY on Multiple Columns

To define a named `PRIMARY KEY` constraint on multiple columns, use the following SQL syntax:

Tutorials Exercises W3Schools app iOS & Android NEW Sign In
. CSS JAVASCRIPT SQL PYTHON JAVA PHP C C++ C# AWS

Note: When using `ALTER TABLE` to add a primary key, the primary key column(s) must have been declared with NOT NULL upon creation of the table.

Drop a PRIMARY KEY Constraint

To drop a `PRIMARY KEY` constraint, use the following SQL:

```
ALTER TABLE Persons
DROP PRIMARY KEY;
```

‹ Previous Sign in to track progress Next ›

Tutorials Exercises W3Schools app iOS & Android NEW Sign In

. CSS JAVASCRIPT SQL PYTHON JAVA PHP C C++ C# AWS

Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

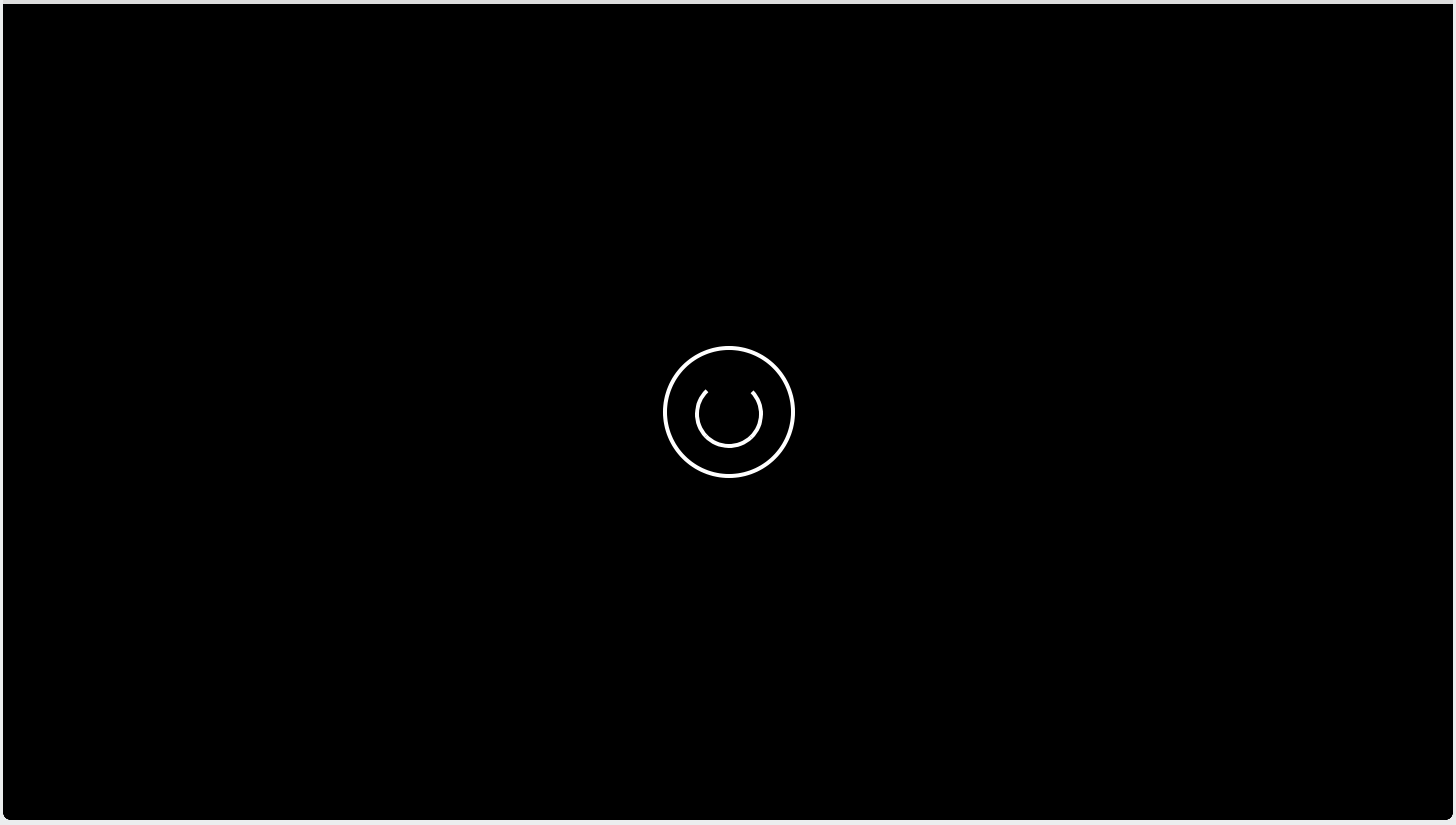
Exam and certificate included.

Get started





REMOVE ADS

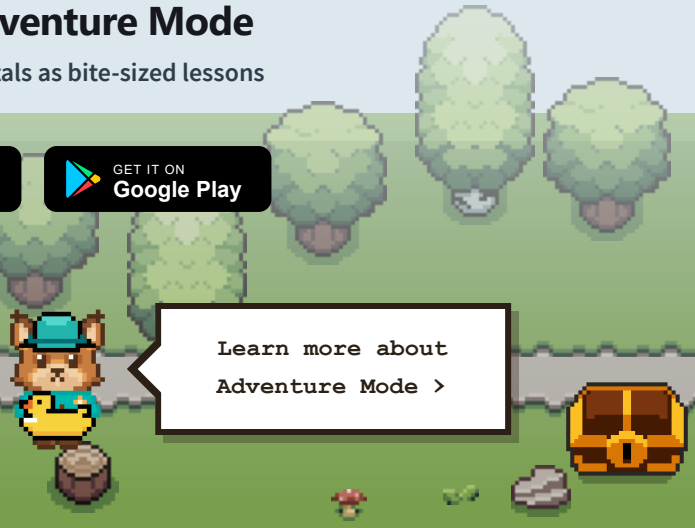


schools Tutorials Exercises W3Schools app iOS & Android NEW Sign In

. CSS JAVASCRIPT SQL PYTHON JAVA PHP C C++ C# AWS

FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play



schools PLUS SPACES

schools Tutorials Exercises W3Schools app iOS & Android NEW Sign In

. CSS JAVASCRIPT SQL PYTHON JAVA PHP C C++ C# AWS

PRACTICE CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

Tutorials Exercises W3Schools app iOS & Android NEW Sign In

. CSS JAVASCRIPT SQL PYTHON JAVA PHP C C++ C# AWS

correctness
of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.